第八讲

限值因变量模型

概述

- 限值因变量(limited dependent variable, LDV):即取值范围明显受到限制的因变量
  - 例：因变量是取1/0的二值变量(是否购买)
  - 例：因变量为非负数(保险理赔次数,购买商品或服务次数)取值等
- 二值因变量的回归建模
  - 线性概率模型、对数单位模型、概率单位模型
- 泊松回归模型及其延展
- 托宾回归模型：截取正态回归模型
  - 不完整观测的建模
- 选择性样本的回归建模：断尾正态回归模型

二值因变量回归建模

二值因变量回归建模

- 二值因变量：因变量只取0/1两个值
- 二值响应模型(binary response models):用于二值因变量的回归建模
  - 目标：响应概率建模

响应概率 $\mathrm{P}(y = 1|\boldsymbol{x}) = \mathrm{P}(y = 1|x_1, x_2, \ldots, x_k),$

  - 给定自变量x条件下，对因变量y等于1的概率建模
- 二值响应模型主要包括
  - 线性概率模型
  - 概率单位模型
  - 对数单位模型

简单二值响应模型
---线性概率模型

- 线性概率模型(Linear Probability Model, LPM)(7.5节)
  - 一种最简单的二值响应模型
  - 源于一般线回归模型

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u,$$

$$\mathrm{E}(y|\boldsymbol{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k. \qquad \mathrm{P}(y = 1|\boldsymbol{x}) = \mathrm{E}(y|\boldsymbol{x})$$

  - 模型形式为： $\mathrm{P}(y = 1|\boldsymbol{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k,$
  - 系数的含义：β_j度量了x_j的变化导致$\mathrm{P}(y = 1)$的变化：

$$\Delta\mathrm{P}(y = 1|\boldsymbol{x}) = \beta_j \Delta x_j.$$

- 线性概率模型示例:分析受教育程度对妇女外出工作的影响(MROZ.txt)
  - 是否工作，其他收入来源，受教育程度，工作经历，年龄，年龄小于6岁的子女数，年龄在6-18岁之间的子女数

```
lm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE +
    KIDSLT6 + KIDSGE6, data = Data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.93432 -0.37526  0.08833  0.34404  0.99417

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.5855192  0.1541780   3.798 0.000158 ***
NWIFEINC    -0.0034052  0.0014485  -2.351 0.018991 *
EDUC         0.0379953  0.0073760   5.151 3.32e-07 ***
EXPER        0.0394924  0.0056727   6.962 7.38e-12 ***
EXPERSQ     -0.0005963  0.0001848  -3.227 0.001306 **
AGE         -0.0160908  0.0024847  -6.476 1.71e-10 ***
KIDSLT6     -0.2618105  0.0335058  -7.814 1.89e-14 ***
KIDSGE6      0.0130122  0.0131960   0.986 0.324415
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4271 on 745 degrees of freedom
Multiple R-squared:  0.2642,    Adjusted R-squared:  0.2573
F-statistic: 38.22 on 7 and 745 DF,  p-value: < 2.2e-16
```

- 一般规定：概率预测值小于0.5，类别预测值为0，否则为1

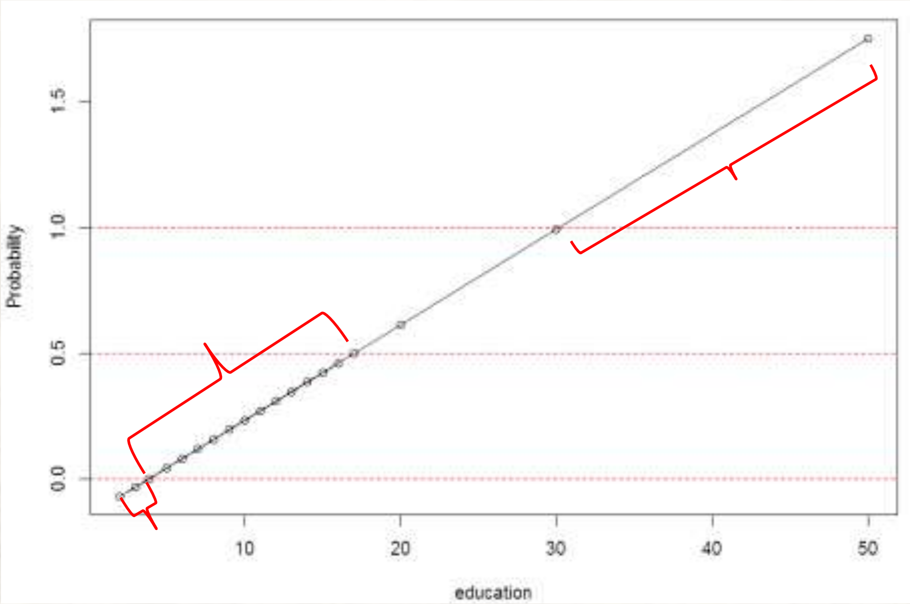
```
. regress INLF EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC AGE

      Source |       SS       df       MS              Number of obs =     753
-------------+------------------------------           F(7, 745)     =   38.22
       Model |  48.8080578     7  6.97257969           Prob > F      =  0.0000
    Residual |  135.919698   745  .182442547           R-squared     =  0.2642
-------------+------------------------------           Adj R-squared =  0.2573
       Total |  184.727756   752  .245648611           Root MSE      =  .42713

------------------------------------------------------------------------------
        INLF |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        EDUC |   .0379953    .007376     5.15   0.000      .023515    .0524756
       EXPER |   .0394924   .0056727     6.96   0.000     .0283561    .0506287
     EXPERSQ |  -.0005963   .0001848    -3.23   0.001    -.0009591   -.0002335
     KIDSGE6 |   .0130122    .013196     0.99   0.324    -.0128935    .0389179
     KIDSLT6 |  -.2618105   .0335058    -7.81   0.000    -.3275875   -.1960335
    NWIFEINC |  -.0034052   .0014485    -2.35   0.019    -.0062488   -.0005616
         AGE |  -.0160908   .0024847    -6.48   0.000    -.0209686    -.011213
       _cons |   .5855192    .154178     3.80   0.000     .2828442    .8881943
------------------------------------------------------------------------------
```

- 线性概率模型的问题：
  - 无法保证模型给出的概率预测值限制在0-1范围内
    - 前例中，当：

$nwifeinc = 50, exper = 5, age = 30, kidslt6 = 1, \text{ and } kidsge6 = 0$



•educ小于3.84时概率为负数
•3.84<educ<=17时概率值小于0.5
•educ>30时概率值大于1

- 本例取值的特殊性，不会出现不合理情况，但仍是一个不能忽视的问题

```
> summary(Data$EDUC)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   5.00   12.00   12.00   12.29   13.00   17.00
```

- 线性概率模型的问题：
  - 线性概率模型反映了概率与自变量取值之间的线性关系，而非一般经验上的非线性关系
    - 例：年收入和购买某品牌奢侈品的概率
  - 违背高斯—马尔科夫假定:误差项的方差，当y为0/1二值变量时：

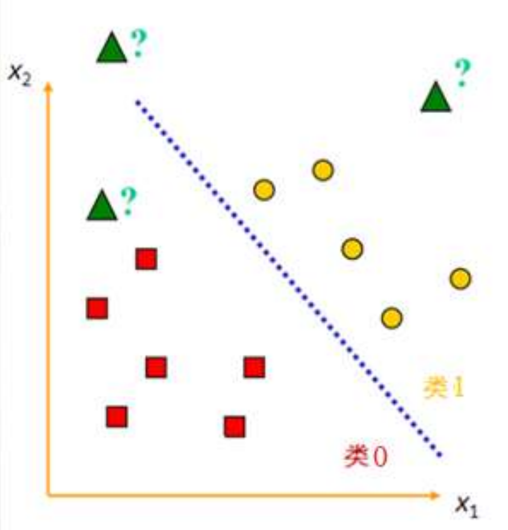
$$\mathrm{Var}(y|\boldsymbol{x}) = p(\boldsymbol{x})[1 - p(\boldsymbol{x})],$$

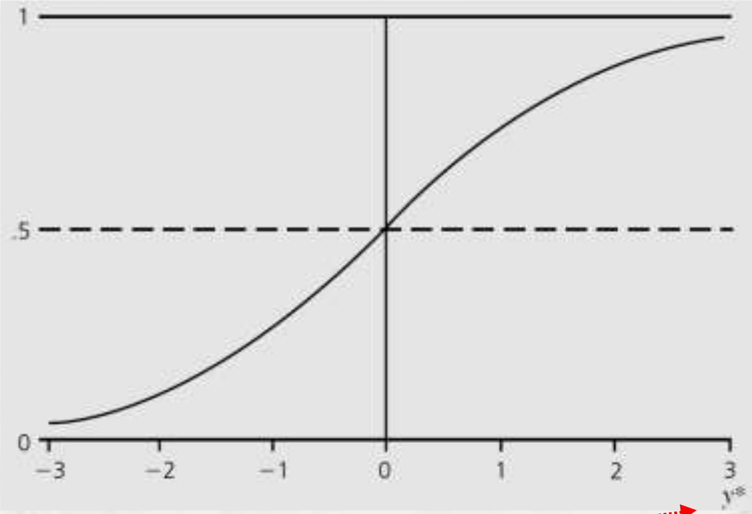
$$p(\boldsymbol{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k.$$

    - 除非概率与任何一个x都不相关，否则一定无法满足同方差性假定

复杂二值响应模型

- 目的：





- 直线方程：

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0, \boldsymbol{x\beta} = 0$$

- $\mathrm{P}(y = 1|\boldsymbol{x})$为点与直线相对位置关系的非线性函数

$$\mathrm{P}(y = 1|\boldsymbol{x}) = G(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k) = G(\beta_0 + \boldsymbol{x\beta}),$$

  - 非线性函数值越大，$P(y = 1|\boldsymbol{x})$越大,判为1类的置信度越大；反之

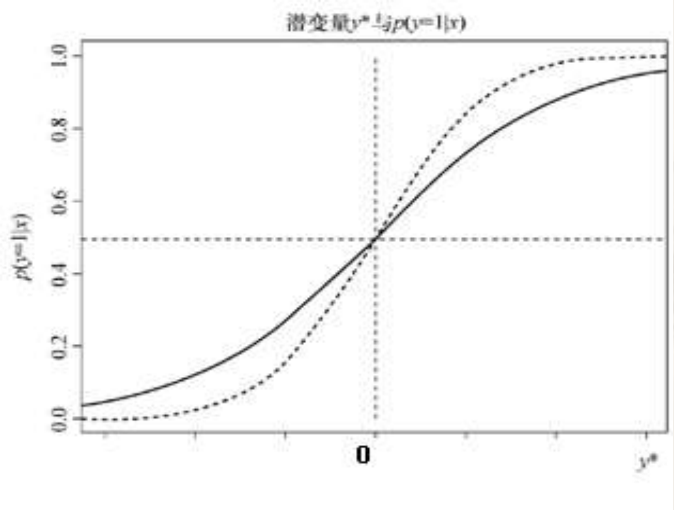
- 目的:克服线性概率模型的局限性
- 策略:以满足高斯马尔可夫假设的潜变量模型为基础
  - 显变量:值能被观测到的变量(记为y，取1或0)
  - 潜变量:值无法被观测到的变量(记为y^*，如购买带来的效益，等)
    - 潜变量模型为一般线性模型,潜变量取值不受限制

$y^* = \beta_0 + \boldsymbol{x\beta} + e,\ y = 1[y^* > 0]$. 示性函数　满足高斯马尔可夫假设

- $\mathrm{P}(y=1|\boldsymbol{x})$是$y^*$的非线性函数$G$，随$y^*$的增加呈非线性增加

$$\mathrm{P}(y=1|\boldsymbol{x}) = G(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k) = G(\beta_0 + \boldsymbol{x\beta}),$$

$$y_i^* > 0 : \mathrm{P}(y_i = 1 | \mathbf{x}_i) > 0.5, \hat{y}_i = 1$$
$$y_i^* < 0 : \mathrm{P}(y_i = 1 | \mathbf{x}_i) < 0.5, \hat{y}_i = 0$$



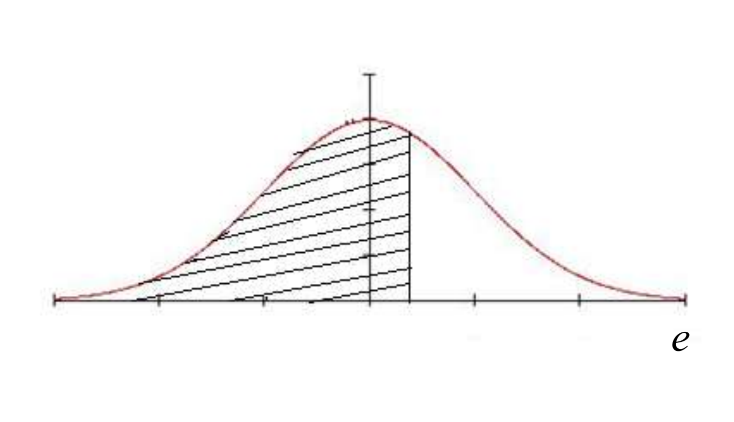
$G()$的函数值在0-1之间,一般可选用某种累积分布函数(CDF)

复杂二值响应模型

- 因： $y^* = \beta_0 + \boldsymbol{x\beta} + e,\ y = 1[y^* > 0].$

$$P(y = 1|\boldsymbol{x}) = G(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k) = G(\beta_0 + \boldsymbol{x\beta}),$$

$$\begin{aligned} P(y = 1|\boldsymbol{x}) &= P(y^* > 0|\boldsymbol{x}) = P[e > -(\beta_0 + \boldsymbol{x\beta})|\boldsymbol{x}] \\ &= 1 - G[-(\beta_0 + \boldsymbol{x\beta})] = G(\beta_0 + \boldsymbol{x\beta}), \end{aligned}$$



利用密度函数关于原点对称的性质

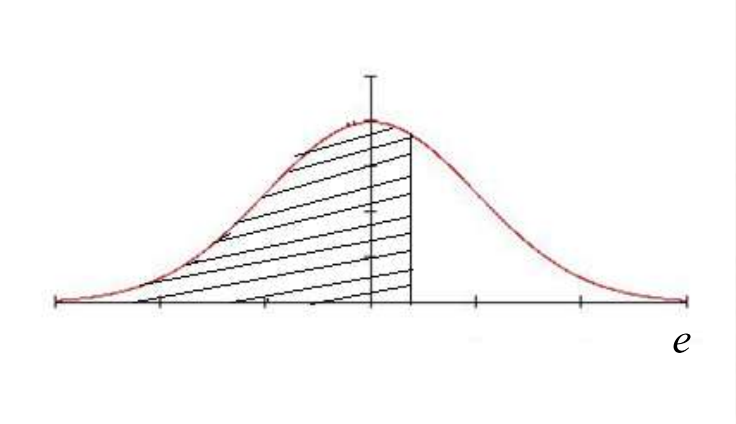
- $G()$的具体形式取决于e的分布
  - 若假设：e是服从N(0,1)标准正态分布的随机变量,模型称为概率单位模型(Probability unit, Probit)
  - 若假设：e是服从逻辑斯蒂分布的随机变量，模型称为对数单位模型(Logit)

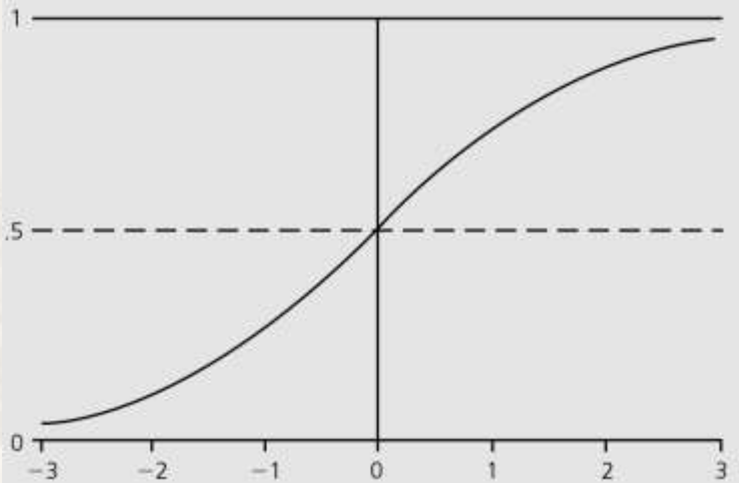
- Probit模型

$$P(y=1|\boldsymbol{x}) = P(y^* > 0|\boldsymbol{x}) = P[e > -(\beta_0 + \boldsymbol{x\beta})|\boldsymbol{x}]$$
$$= 1 - G[-(\beta_0 + \boldsymbol{x\beta})] = G(\beta_0 + \boldsymbol{x\beta}),$$

复杂二值响应模型

- e服从标准正态分布(潜变量模型的误差项方差为1),概率密度函数PDF，记为$\phi(e)$
- e的累计分布函数为标准正态分布的CDF，记为$\boldsymbol{\Phi}(z) \equiv \int_{-\infty}^{z} \phi(e)\mathrm{d}e$





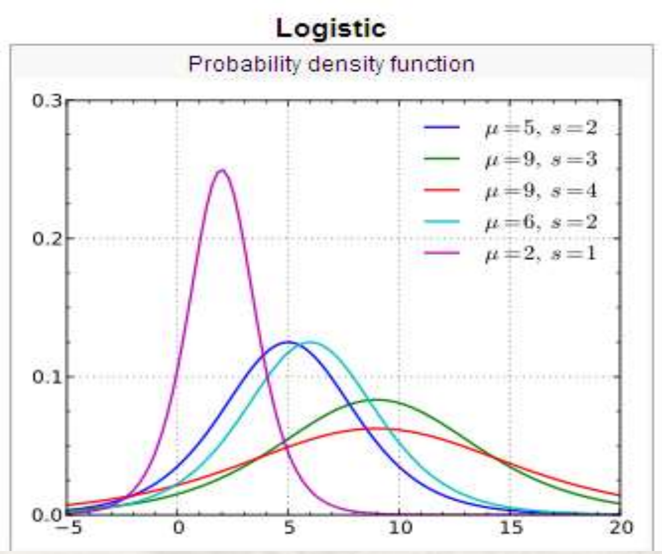
- $P(y=1|\boldsymbol{x}) = G(\beta_0 + \boldsymbol{x\beta}) = \boldsymbol{\Phi}(\beta_0 + \boldsymbol{x\beta})$,等于PDF曲线下的阴影面积

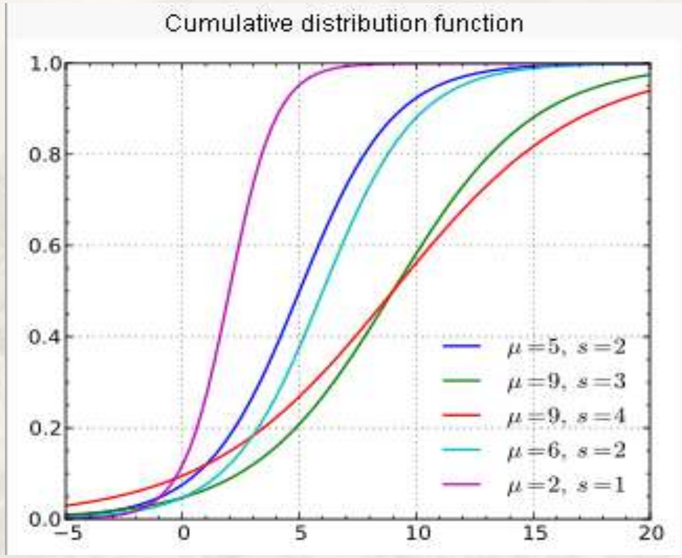
■ 对数单位Logit模型

$$
\begin{aligned}
\mathrm{P}(y=1|\boldsymbol{x}) &= \mathrm{P}(y^*>0|\boldsymbol{x}) = \mathrm{P}[e>-(\beta_0+\boldsymbol{x\beta})|\boldsymbol{x}] \\
&= 1-G[-(\beta_0+\boldsymbol{x\beta})] = G(\beta_0+\boldsymbol{x\beta}),
\end{aligned}
$$

复杂二值响应模型

■ e是逻辑斯蒂随机变量(潜变量模型的误差项方差为$\pi^2/3$)，概率密度函数PDF：





$$
f(x;\mu,s) = \frac{e^{-\frac{x-\mu}{s}}}{s\left(1+e^{-\frac{x-\mu}{s}}\right)^2}
$$

```
> set.seed(12345)   #logistic分布中的标准差
> s<-NULL
> for(i in 1:100)
+ s<-c(s,sd(rlogis(1000,0,1)))
> mean(s)
[1] 1.811961
```

$\mu = 0, s = 1$

•s是标准差σ的比例值

$$
s = q\,\sigma
$$
$$
q = \sqrt{3}/\pi = 0.551328895
$$

•s=1时$\sigma = 1.8 = \frac{\pi}{\sqrt{3}}$

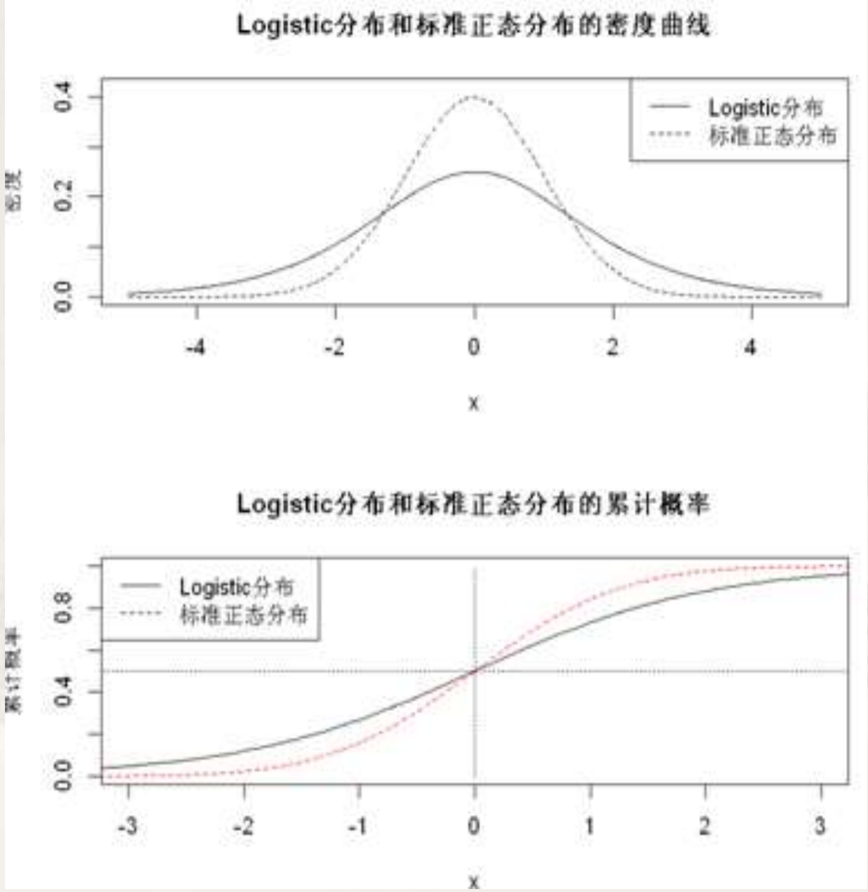
■ e的CDF：

$$
F(x;\mu,s) = \frac{1}{1+e^{-\frac{x-\mu}{s}}}
$$

$$
\mathrm{P}(y=1|\boldsymbol{x}) = G(\beta_0+\boldsymbol{x\beta}) = F(\beta_0+\boldsymbol{x\beta};\mu=0,s=1) = \frac{1}{1+e^{-(\beta_0+\boldsymbol{x\beta})}}
$$

复杂二值响应模型

- Probit模型和Logit模型
  - Probit的$P(y=1|\boldsymbol{x})$变化比Logit(实线)更快些



复杂二值响应模型的参数估计

- 极大似然估计:在所有可能的参数$\boldsymbol{\Theta}$值集合中找到使样本有最大出现概率时的θ

$$\begin{aligned}\mathrm{P}(y=1|\boldsymbol{x}) &= \mathrm{P}(y^*>0|\boldsymbol{x}) = \mathrm{P}[e>-(\beta_0+\boldsymbol{x\beta})|\boldsymbol{x}] \\ &= 1-G[-(\beta_0+\boldsymbol{x\beta})] = G(\beta_0+\boldsymbol{x\beta}),\end{aligned}$$

  - 概率函数:

$$\mathrm{P}(y_i=1|\boldsymbol{x}_i;\boldsymbol{\beta}) = G(\boldsymbol{x}_i\boldsymbol{\beta}) \quad \mathrm{P}(y_i=0|\boldsymbol{x}_i;\boldsymbol{\beta}) = 1-G(\boldsymbol{x}_i\boldsymbol{\beta})$$

$$\mathrm{P}(y_i|\boldsymbol{x}_i;\boldsymbol{\beta}) = [G(\boldsymbol{x}_i\boldsymbol{\beta})^{y_i}][1-G(\boldsymbol{x}_i\boldsymbol{\beta})^{1-y_i}],(y_i=1,0)$$

  - 构造似然函数(观测到n个独立的样本观测1或0):

$$L(y|\boldsymbol{x};\boldsymbol{\beta}) = \prod_{i=1}^{n}[G(\boldsymbol{x}_i\boldsymbol{\beta})^{y_i}][1-G(\boldsymbol{x}_i\boldsymbol{\beta})^{1-y_i}]$$

复杂二值响应模型的参数估计

- 构造对数似然函数

$$L(y|\boldsymbol{x};\boldsymbol{\beta})=\prod_{i=1}^{n}[G(\boldsymbol{x}_i\boldsymbol{\beta})^{y_i}][1-G(\boldsymbol{x}_i\boldsymbol{\beta})^{1-y_i}]$$

  - 对数似然函数

$$LL(y|\boldsymbol{x};\boldsymbol{\beta})=\sum_{i=1}^{n}[y_i\ln G(\boldsymbol{x}_i\boldsymbol{\beta})+(1-y_i)\ln(1-G(\boldsymbol{x}_i\boldsymbol{\beta})]$$

$$LL(y|\boldsymbol{x};\boldsymbol{\beta})=\sum_{i=1}^{n}[y_i\ln\frac{G(\boldsymbol{x}_i\boldsymbol{\beta})}{1-G(\boldsymbol{x}_i\boldsymbol{\beta})}+\ln(1-G(\boldsymbol{x}_i\boldsymbol{\beta}))]$$

    - Logit模型和Probit模型，对数似然函数中的G不同

- 最大化对数似然函数，得到$\boldsymbol{\beta}$的极大似然估计

■ Probit模型示例：分析受教育程度对职业妇女外出工作的影响(MROZ.txt)

```
> Data<-read.table(file="mroz.txt",header=TRUE,sep="")
> plot(factor(INLF)~EDUC,data=Data,ylevels=2:1)
> mroz_probit<-glm(INLF~NWIFEINC+EDUC+EXPER+EXPERSQ+AGE+KIDSLT6+KIDSGE6,data=Data,
+     family=binomial(link="probit"))
> summary(mroz_probit)

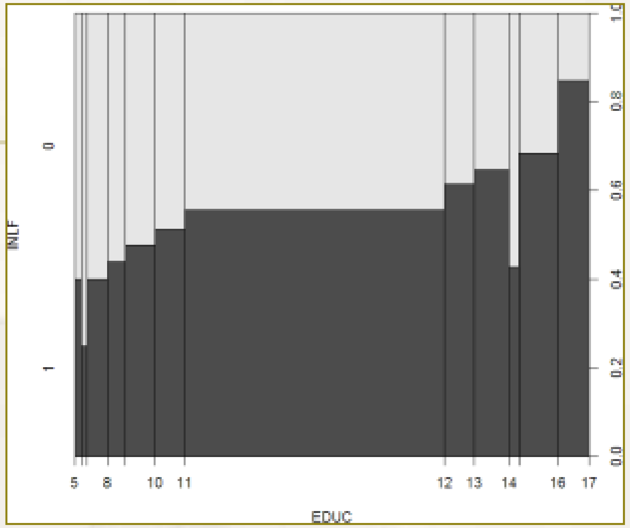
Call:
glm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE +
    KIDSLT6 + KIDSGE6, family = binomial(link = "probit"), data = Data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.2156  -0.9151   0.4315   0.8653   2.4553

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.2700736  0.5080782   0.532  0.59503
NWIFEINC    -0.0120236  0.0049392  -2.434  0.01492 *
EDUC         0.1309040  0.0253987   5.154 2.55e-07 ***
EXPER        0.1233472  0.0187587   6.575 4.85e-11 ***
EXPERSQ     -0.0018871  0.0005999  -3.145  0.00166 **
AGE         -0.0528524  0.0084624  -6.246 4.22e-10 ***
KIDSLT6     -0.8683247  0.1183773  -7.335 2.21e-13 ***
KIDSGE6      0.0360056  0.0440303   0.818  0.41350
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.7  on 752  degrees of freedom
Residual deviance:  802.6  on 745  degrees of freedom
AIC: 818.6
```




```
. probit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC,nolog

Probit regression                               Number of obs   =        753
                                                LR chi2(7)      =     227.14
                                                Prob > chi2     =     0.0000
Log likelihood = -401.30219                     Pseudo R2       =     0.2206

------------------------------------------------------------------------------
        INLF |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         AGE |  -.0528527   .0084772    -6.23   0.000    -.0694678   -.0362376
        EDUC |   .1309047   .0252542     5.18   0.000     .0814074     .180402
       EXPER |   .1233476   .0187164     6.59   0.000     .0866641    .1600311
     EXPERSQ |  -.0018871      .0006    -3.15   0.002     -.003063   -.0007111
     KIDSGE6 |    .036005   .0434768     0.83   0.408     -.049208    .1212179
     KIDSLT6 |  -.8683285   .1185223    -7.33   0.000    -1.100628    -.636029
    NWIFEINC |  -.0120237   .0048398    -2.48   0.013    -.0215096   -.0025378
       _cons |   .2700768    .508593     0.53   0.595    -.7267473    1.266901
------------------------------------------------------------------------------
```

- Probit模型的异方差问题

$$y^* = \beta_0 + \boldsymbol{x\beta} + e, y = 1[y^* > 0],$$

$$P(y = 1|\boldsymbol{x}) = G(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k) = G(\beta_0 + \boldsymbol{x\beta}),$$

  - 同方差假定：

$$P(y_i = 1|\boldsymbol{x}_i) = G\left(\frac{\boldsymbol{x}_i\boldsymbol{\beta}}{\sigma}\right), \sigma^2 = Var(e) = 1 \qquad P(y_i = 1|\boldsymbol{x}_i) = G\left(\frac{\boldsymbol{x}_i\boldsymbol{\beta}}{\sigma_i}\right), \sigma_i^2 = Var(e_i)$$

```
. hetprobit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC,het(AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC) nolog

Heteroskedastic probit model                    Number of obs     =        753
                                                Zero outcomes     =        325
                                                Nonzero outcomes  =        428

                                                Wald chi2(7)      =       1.74
Log likelihood = -397.1186                      Prob > chi2       =     0.9727
```

异方差下的估计结果

INLF	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
INLF						
AGE	-.0754333	.0591184	-1.28	0.202	-.1913032	.0404366
EDUC	.2246601	.1959389	1.15	0.252	-.1593731	.6086933
EXPER	.2080328	.1799471	1.16	0.248	-.144657	.5607226
EXPERSQ	-.0032365	.0032247	-1.00	0.316	-.0095568	.0030838
KIDSGE6	.1304411	.1703058	0.77	0.444	-.2033522	.4642344
KIDSLT6	-1.425694	1.176554	-1.21	0.226	-3.731697	.88031
NWIFEINC	-.0247571	.0237092	-1.04	0.296	-.0712262	.021712
_cons	-.2565155	1.019266	-0.25	0.801	-2.25424	1.741209
lnsigma						
AGE	.0158777	.0135473	1.17	0.241	-.0106746	.04243
EDUC	-.0445655	.0346394	-1.29	0.198	-.1124575	.0233264
EXPER	-.0009441	.0320553	-0.03	0.977	-.0637713	.0618831
EXPERSQ	.0001604	.0013025	0.12	0.902	-.0023925	.0027132
KIDSGE6	.1262551	.0738244	1.71	0.087	-.0184381	.2709482
KIDSLT6	.138953	.1901259	0.73	0.465	-.2336869	.5115929
NWIFEINC	.0076199	.0071436	1.07	0.286	-.0063813	.0216212

```
LR test of lnsigma=0: chi2(7) = 8.37                  Prob > chi2 = 0.3013
```

$$\log(\sigma^2) = \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k$$

- 原假设：$\delta_1 = \delta_2 = \ldots = \delta_k = 0$

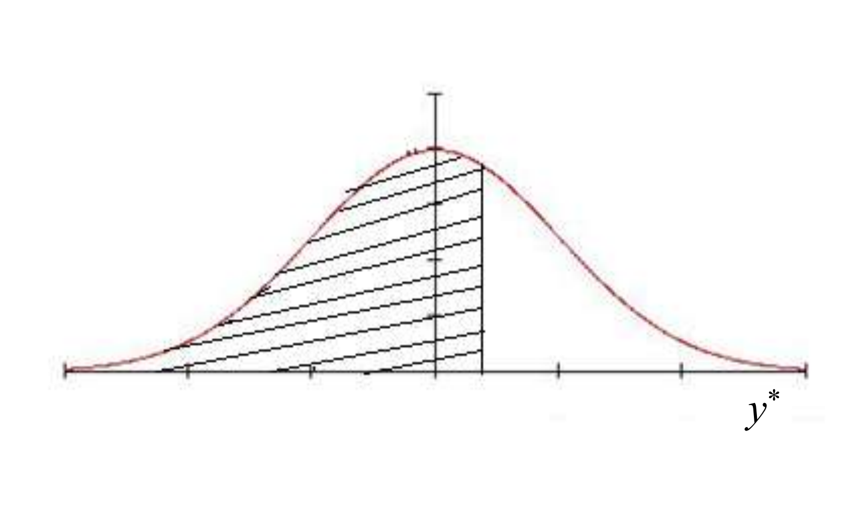
无法拒绝原假设，同方差假定满足

解释变量的偏效应

- 复杂二值响应模型中的系数$\boldsymbol{\beta}$

$$y^* = \beta_0 + \boldsymbol{x\beta} + e, y = 1[y^* > 0].$$

- β_j表示x_j变化一个单位引起的潜变量的变化，含义不直观



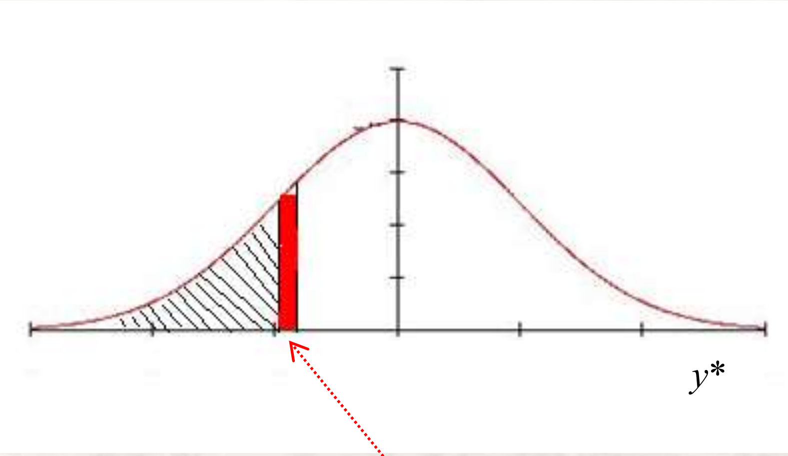
关心图中阴影面积(响应概率P)的变化

解释变量的偏效应

- 关心：数值型变量x_j变化一个单位引起的响应概率P的变化
  - x_j取值大致连续时，x_j变化极小时响应概率的变化近似为图中红色部分的面积

比例因子

$$\Delta\hat{\mathrm{P}}(y=1|\boldsymbol{x}) \approx [\underbrace{g(\hat{\beta}_0+\boldsymbol{x}\hat{\boldsymbol{\beta}})\hat{\beta}_j}]\Delta x_j,$$



•x_j的偏效应与β_j有关

•x_j的偏效应与比例因子(这里为概率密度函数)有关

•比例因子严格为正，偏效应的方向仅取决于β_j的正负

•x_j的偏效应与x_j的当前取值有关，是非线性的

$$\Delta \mathrm{P} = G(\beta_0+\beta_1 x_1+\beta_1\Delta x_1) - G(\beta_0+\beta_1 x_1)$$
$$\approx g(\beta_0+\beta_1 x_1)\times \beta_1\Delta x_1$$

$$\begin{aligned}\mathrm{P}(y=1|\boldsymbol{x}) &= \mathrm{P}(y^*>0|\boldsymbol{x}) = \mathrm{P}[e>-(\beta_0+\boldsymbol{x\beta})|\boldsymbol{x}] \\ &= 1-G[-(\beta_0+\boldsymbol{x\beta})] = G(\beta_0+\boldsymbol{x\beta}),\end{aligned}$$

$$\frac{\partial p(\mathbf{x})}{\partial x_j} = g(\mathbf{x}\boldsymbol{\beta})\beta_j$$

- 为方便量化x_j的偏效应，计算两种偏效应 $\Delta\hat{P}(y=1|\boldsymbol{x}) \approx [g(\hat{\beta}_0 + \boldsymbol{x}\hat{\boldsymbol{\beta}})\hat{\beta}_j]\Delta x_j,$
  - 平均值上的偏效应(partial effect at the average, PEA)
    - 将$\boldsymbol{x}$以均值代入，计算比例因子： $g(\hat{\beta}_0 + \bar{\boldsymbol{x}}\hat{\boldsymbol{\beta}})$.
    - 乘以β_j得到x_j的偏效应
    - 不足：有时$\boldsymbol{x}$的均值没有意义
  - 平均偏效应(average partial effect, APE)，常用
    - 基于偏效应的平均值
      - 以概率密度均值作为比例因子： $\frac{1}{n}\sum_{i=1}^{n} g(\hat{\beta}_0 + \mathbf{x}_i\hat{\boldsymbol{\beta}})$
      - 乘以β_j得到x_j偏效应
- 可避开比例因子，直接利用回归系数$\boldsymbol{\beta}$对比x_j与x_h的偏效应(比较变量重要性)

$$\frac{\partial p(\mathbf{x})}{\partial x_j} = g(\mathbf{x}\boldsymbol{\beta})\beta_j \qquad \frac{\partial p(\mathbf{x})/\partial x_j}{\partial p(\mathbf{x})/\partial x_h} = \beta_j/\beta_h$$

前例：Probit模型分析受教育程度对职业妇女外出工作的影响(MROZ.txt)

```
> Data<-read.table(file="mroz.txt",header=TRUE,sep="")
> plot(factor(INLF)~EDUC,data=Data,ylevels=2:1)
> mroz_probit<-glm(INLF~NWIFEINC+EDUC+EXPER+EXPERSQ+AGE+KIDSLT6+KIDSGE6,data=Data,
+     family=binomial(link="probit"))
> summary(mroz_probit)

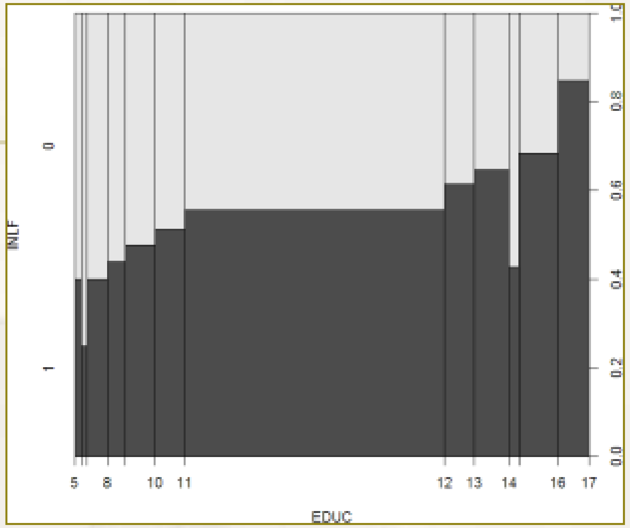
Call:
glm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE +
    KIDSLT6 + KIDSGE6, family = binomial(link = "probit"), data = Data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.2156  -0.9151   0.4315   0.8653   2.4553

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.2700736  0.5080782   0.532  0.59503
NWIFEINC    -0.0120236  0.0049392  -2.434  0.01492 *
EDUC         0.1309040  0.0253987   5.154 2.55e-07 ***
EXPER        0.1233472  0.0187587   6.575 4.85e-11 ***
EXPERSQ     -0.0018871  0.0005999  -3.145  0.00166 **
AGE         -0.0528524  0.0084624  -6.246 4.22e-10 ***
KIDSLT6     -0.8683247  0.1183773  -7.335 2.21e-13 ***
KIDSGE6      0.0360056  0.0440303   0.818  0.41350
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.7  on 752  degrees of freedom
Residual deviance:  802.6  on 745  degrees of freedom
AIC: 818.6
```




```
. probit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC,nolog

Probit regression                               Number of obs   =        753
                                                LR chi2(7)      =     227.14
                                                Prob > chi2     =     0.0000
Log likelihood = -401.30219                     Pseudo R2       =     0.2206

------------------------------------------------------------------------------
        INLF |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         AGE |  -.0528527   .0084772    -6.23   0.000    -.0694678   -.0362376
        EDUC |   .1309047   .0252542     5.18   0.000     .0814074     .180402
       EXPER |   .1233476   .0187164     6.59   0.000     .0866641    .1600311
     EXPERSQ |  -.0018871      .0006    -3.15   0.002     -.003063   -.0007111
     KIDSGE6 |    .036005   .0434768     0.83   0.408     -.049208    .1212179
     KIDSLT6 |  -.8683285   .1185223    -7.33   0.000    -1.100628    -.636029
    NWIFEINC |  -.0120237   .0048398    -2.48   0.013    -.0215096   -.0025378
       _cons |   .2700768    .508593     0.53   0.595    -.7267473    1.266901
------------------------------------------------------------------------------
```

- 前例：Probit模型分析受教育程度对职业妇女外出工作的影响
  - 两种比例因子的计算以及偏效应的对比 $\Delta\hat{P}(y=1|\boldsymbol{x}) \approx [g(\hat{\beta}_0 + \boldsymbol{x}\hat{\boldsymbol{\beta}})\hat{\beta}_j]\Delta x_j,$

```
> ###计算PEA
> avX<-colMeans(Data[,c("NWIFEINC","EDUC","EXPER","EXPERSQ","AGE","KIDSLT6","KIDSGE6")])
> avYstarM<-predict(mroz_probit,newdata=as.data.frame(t(avX)),type="link")
> (PEA<-dnorm(avYstarM)*coef(mroz_probit))
 (Intercept)     NWIFEINC         EDUC        EXPER      EXPERSQ          AGE      KIDSLT6
0.1054852458 -0.0046961882 0.0511284288 0.0481768957 -0.0007370502 -0.0206430891 -0.3391499646 0.
> ##计算APE
> (APE<-mean(dnorm(predict(mroz_probit,newdata=Data,type="link")))*coef(mroz_probit))
 (Intercept)    NWIFEINC        EDUC       EXPER     EXPERSQ         AGE     KIDSLT6     KIDSG
0.081226125 -0.003616176 0.039370095 0.037097345 -0.000567546 -0.015895665 -0.261153464 0.0108288
```

```
. margins, dydx(*)

Average marginal effects                        Number of obs     =        753
Model VCE    : OIM

Expression   : Pr(INLF), predict()
dy/dx w.r.t. : AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC
```

	dy/dx	Delta-method Std. Err.	z	P>|z|	[95% Conf.	Interval]
AGE	-.0157194	.0023808	-6.60	0.000	-.0203856	-.0110532
EDUC	.0394965	.0072947	5.41	0.000	.0251992	.0537939
EXPER	.0367641	.00515	7.14	0.000	.0266702	.046858
EXPERSQ	-.0005633	.0001774	-3.18	0.001	-.0009109	-.0002156
KIDSGE6	.0107348	.013333	0.81	0.421	-.0153974	.0368671
KIDSLT6	-.2577537	.0319416	-8.07	0.000	-.3203581	-.1951492
NWIFEINC	-.0038118	.0014824	-2.57	0.010	-.0067172	-.0009064

```
. margins, dydx(*) atmeans

Conditional marginal effects                    Number of obs     =        753
Model VCE    : OIM

Expression   : Pr(INLF), predict()
dy/dx w.r.t. : AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC
at           : AGE             =    42.53785 (mean)
               EDUC            =    12.28685 (mean)
               EXPER           =    10.63081 (mean)
               EXPERSQ         =    178.0385 (mean)
               KIDSGE6         =    1.353254 (mean)
               KIDSLT6         =    .2377158 (mean)
               NWIFEINC        =    20.12896 (mean)
```

	dy/dx	Delta-method Std. Err.	z	P>|z|	[95% Conf.	Interval]
AGE	-.0206432	.0033079	-6.24	0.000	-.0271265	-.0141598
EDUC	.0511287	.0098592	5.19	0.000	.0318051	.0704523
EXPER	.0481771	.0073278	6.57	0.000	.0338149	.0625392
EXPERSQ	-.0007371	.0002347	-3.14	0.002	-.001197	-.0002771
KIDSGE6	.0140628	.0169852	0.83	0.408	-.0192275	.0473531
KIDSLT6	-.3391514	.0463581	-7.32	0.000	-.4300117	-.2482911
NWIFEINC	-.0046962	.0018903	-2.48	0.013	-.0084012	-.0009913

- 妇女受教育程度EDUC增加一年
  - PEA估计：出去工作的概率平均提高0.05；APE估计：出去工作的概率平均提高0.04

- Logit模型示例：分析受教育程度对职业妇女外出工作的影响(MROZ.txt)

```
> mroz_logit<-glm(INLF~NWIFEINC+EDUC+EXPER+EXPERSQ+AGE+KIDSLT6+KIDSGE6,data=Data,
+    family=binomial(link="logit"))
> summary(mroz_logit)

Call:
glm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE +
    KIDSLT6 + KIDSGE6, family = binomial(link = "logit"), data = Data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.1770  -0.9063   0.4473   0.8561   2.4032

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.425452   0.860365   0.495  0.62095
NWIFEINC    -0.021345   0.008421  -2.535  0.01126 *
EDUC         0.221170   0.043439   5.091 3.55e-07 ***
EXPER        0.205870   0.032057   6.422 1.34e-10 ***
EXPERSQ     -0.003154   0.001016  -3.104  0.00191 **
AGE         -0.088024   0.014573  -6.040 1.54e-09 ***
KIDSLT6     -1.443354   0.203583  -7.090 1.34e-12 ***
KIDSGE6      0.060112   0.074789   0.804  0.42154
---
```

```
. logit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC, nolog

Logistic regression                               Number of obs   =        753
                                                  LR chi2(7)      =     226.22
                                                  Prob > chi2     =     0.0000
Log likelihood = -401.76515                       Pseudo R2       =     0.2197

------------------------------------------------------------------------------
        INLF |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         AGE |  -.0880244    .014573    -6.04   0.000     -.116587   -.0594618
        EDUC |   .2211704   .0434396     5.09   0.000     .1360303    .3063105
       EXPER |   .2058695   .0320569     6.42   0.000     .1430391    .2686999
     EXPERSQ |  -.0031541   .0010161    -3.10   0.002    -.0051456   -.0011626
     KIDSGE6 |   .0601122   .0747897     0.80   0.422     -.086473    .2066974
     KIDSLT6 |  -1.443354   .2035849    -7.09   0.000    -1.842373   -1.044335
    NWIFEINC |  -.0213452   .0084214    -2.53   0.011    -.0378509   -.0048394
       _cons |   .4254524   .8603697     0.49   0.621    -1.260841    2.111746
------------------------------------------------------------------------------
```

- Logit模型示例：分析受教育程度对职业妇女外出工作的影响(MROZ.txt)

```
. margins, dydx(*) atmeans  //PEA

Conditional marginal effects                    Number of obs     =        753
Model VCE    : OIM

Expression   : Pr(INLF), predict()
dy/dx w.r.t. : AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC
at           : AGE             =    42.53785 (mean)
               EDUC            =    12.28685 (mean)
               EXPER           =    10.63081 (mean)
               EXPERSQ         =    178.0385 (mean)
               KIDSGE6         =    1.353254 (mean)
               KIDSLT6         =    .2377158 (mean)
               NWIFEINC        =    20.12896 (mean)
```

	dy/dx	Delta-method Std. Err.	z	P>|z|	[95% Conf.	Interval]
AGE	-.021403	.0035398	-6.05	0.000	-.0283408	-.0144652
EDUC	.0537773	.0105608	5.09	0.000	.0330785	.0744761
EXPER	.0500569	.0078247	6.40	0.000	.0347209	.065393
EXPERSQ	-.0007669	.0002477	-3.10	0.002	-.0012524	-.0002815
KIDSGE6	.0146162	.0181884	0.80	0.422	-.0210324	.0502649
KIDSLT6	-.3509498	.0496395	-7.07	0.000	-.4482414	-.2536583
NWIFEINC	-.0051901	.0020482	-2.53	0.011	-.0092045	-.0011756

```
. margins, dydx(*)  //APE

Average marginal effects                        Number of obs     =        753
Model VCE    : OIM

Expression   : Pr(INLF), predict()
dy/dx w.r.t. : AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC
```

	dy/dx	Delta-method Std. Err.	z	P>|z|	[95% Conf.	Interval]
AGE	-.0157194	.0023808	-6.60	0.000	-.0203856	-.0110532
EDUC	.0394965	.0072947	5.41	0.000	.0251992	.0537939
EXPER	.0367641	.00515	7.14	0.000	.0266702	.046858
EXPERSQ	-.0005633	.0001774	-3.18	0.001	-.0009109	-.0002156
KIDSGE6	.0107348	.013333	0.81	0.421	-.0153974	.0368671
KIDSLT6	-.2577537	.0319416	-8.07	0.000	-.3203581	-.1951492
NWIFEINC	-.0038118	.0014824	-2.57	0.010	-.0067172	-.0009064

```
. margins, dydx(*) at (AGE = 30)

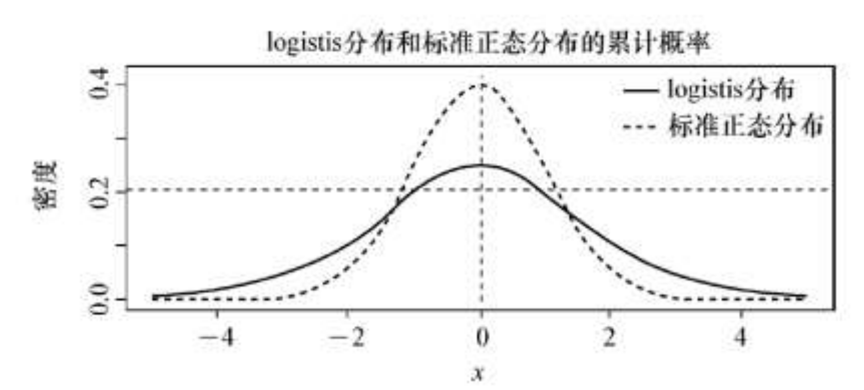
Average marginal effects                        Number of obs     =        753
Model VCE    : OIM

Expression   : Pr(INLF), predict()
dy/dx w.r.t. : AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC
at           : AGE             =          30
```

	dy/dx	Delta-method Std. Err.	z	P>|z|	[95% Conf.	Interval]
AGE	-.0122137	.0012395	-9.85	0.000	-.014643	-.0097843
EDUC	.0306881	.0063633	4.82	0.000	.0182163	.04316
EXPER	.0285651	.0044799	6.38	0.000	.0197846	.0373456
EXPERSQ	-.0004376	.0001416	-3.09	0.002	-.0007153	-.00016
KIDSGE6	.0083408	.01059	0.79	0.431	-.0124152	.0290967
KIDSLT6	-.2002702	.0217035	-9.23	0.000	-.2428082	-.1577322
NWIFEINC	-.0029617	.0012049	-2.46	0.014	-.0053232	-.0006002

- 妇女受教育程度EDUC增加一年
  - PEA估计：出去工作的概率平均提高0.05
  - APE估计：出去工作的概率平均提高0.04
- 偏效应估计结果与Probit近似相等

- Probit和Logit模型的选择问题
  - 偏效应估计的对比：除β_j还需计算比例因子。对比两个模型的PEA或APE(繁琐)
  - 粗略做法：$\Delta\hat{P}(y=1|\boldsymbol{x}) \approx [g(\hat{\beta}_0 + \boldsymbol{x}\hat{\boldsymbol{\beta}})\hat{\beta}_j]\Delta x_j,$
    - 两模型密度函数不同，比例因子对偏效应的贡献不同
    - “基准”：$y^* = 0$时的密度 $\mathbf{x}\boldsymbol{\beta} = 0$



- Probit模型：$\phi(z) = (2\pi)^{-1/2}\exp(-z^2/2).$
  g(0)=ϕ(0)=1/sqrt(2π)=0.4
- Logit模型：$\mu = 1, s = 0, g(0) = 0.25$

$$f(x;\mu,s) = \frac{e^{-\frac{x-\mu}{s}}}{s\left(1+e^{-\frac{x-\mu}{s}}\right)^2}$$

- 若两模型有大致相同的偏效应估计，则两个β的关系：

$$g(\text{Probit})\breve{\beta}_{probit} \approx g(\text{Logit})\hat{\beta}_{Logit}$$

$$\frac{g(\text{Probit})}{g(\text{Logit})} = \frac{0.4}{0.25} \approx \frac{\hat{\beta}_{Logit}}{\breve{\beta}_{probit}}$$

概率单位模型的$\boldsymbol{\beta}$乘以0.4/0.25=1.6，应近似等于对数单位模型的$\boldsymbol{\beta}$

- Probit和Logit模型的选择问题:分析影响职业妇女外出工作的因素(MROZ.txt)

```
glm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE +
    KIDSLT6 + KIDSGE6, family = binomial(link = "probit"), data = Data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.2156  -0.9151   0.4315   0.8653   2.4553

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.2700736  0.5080782   0.532  0.59503
NWIFEINC    -0.0120236  0.0049392  -2.434  0.01492 *
EDUC         0.1309040  0.0253987   5.154 2.55e-07 ***
EXPER        0.1233472  0.0187587   6.575 4.85e-11 ***
EXPERSQ     -0.0018871  0.0005999  -3.145  0.00166 **
AGE         -0.0528524  0.0084624  -6.246 4.22e-10 ***
KIDSLT6     -0.8683247  0.1183773  -7.335 2.21e-13 ***
KIDSGE6      0.0360056  0.0440303   0.818  0.41350
```

```
> mroz_logit<-glm(INLF~NWIFEINC+EDUC+EXPER+EXPERSQ+AGE+KIDSLT6+KIDSGE6,data=Data,
+    family=binomial(link="logit"))
> summary(mroz_logit)

Call:
glm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE +
    KIDSLT6 + KIDSGE6, family = binomial(link = "logit"), data = Data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.1770  -0.9063   0.4473   0.8561   2.4032

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.425452   0.860365   0.495  0.62095
NWIFEINC    -0.021345   0.008421  -2.535  0.01126 *
EDUC         0.221170   0.043439   5.091 3.55e-07 ***
EXPER        0.205870   0.032057   6.422 1.34e-10 ***
EXPERSQ     -0.003154   0.001016  -3.104  0.00191 **
AGE         -0.088024   0.014573  -6.040 1.54e-09 ***
KIDSLT6     -1.443354   0.203583  -7.090 1.34e-12 ***
KIDSGE6      0.060112   0.074789   0.804  0.42154
```

$$g(\text{Probit})\breve{\beta}_j \Leftrightarrow g(\text{Logit})\hat{\beta}_j$$

$$0.4\breve{\beta}_j \Leftrightarrow 0.25\hat{\beta}_j$$

$$0.4 * 0.13 = 0.052$$

$$0.25 * 0.22 = 0.055$$

两模型给出了相近的偏效应估计

- Probit和Logit模型的选择问题
  - Logit模型$\boldsymbol{\beta}$较Probit模型，实际含义更直观
  - 其他问题中Probit模型更好(以后讲)

$$\mathrm{P}(y=1|\boldsymbol{x})=G(\beta_0+\boldsymbol{x\beta})=\frac{1}{1+e^{-(\beta_0+\boldsymbol{x\beta})}}$$

概率与自变量呈非线性

$$\mathrm{P}=\frac{\exp(\beta_0+\mathbf{x\beta})}{1+\exp(\beta_0+\mathbf{x\beta})} \qquad \mathrm{P}[1+\exp(\beta_0+\mathbf{x\beta})]=\exp(\beta_0+\mathbf{x\beta})$$

$$\frac{\mathrm{P}}{1-\mathrm{P}}=\exp(\beta_0+\mathbf{x\beta})$$

$$\ln\left(\frac{\mathrm{P}}{1-\mathrm{P}}\right)=\beta_0+\mathbf{x\beta}$$

Logit P的取值范围满足一般线性模型的要求

Logit变换(一种连接函数)

$$\underbrace{Logit\ \mathrm{P}}_{y^*}=\beta_0+\mathbf{x\beta}$$

- Logit模型中： $\ln(\frac{P}{1-P})=\beta_0+\mathbf{x\beta}$ β的实际意义仍不清晰
  - 于是：

$$\Omega=\frac{P}{1-P}=\exp(\beta_0+\mathbf{x\beta})$$

优势(odds)

$$\Omega=\exp(\beta_0+\mathbf{x\beta})$$

例：有x_1 x_2两个自变量，x_1是二值解释变量。当x_1从0变化到1时：

$$\Omega=\exp(\beta_0+\beta_2x_2)$$

$$\Omega^*=\exp(\beta_1+\beta_0+\beta_2x_2)=\Omega\exp(\beta_1)$$

优势比

$$\frac{\Omega^*}{\Omega}=\exp(\beta_1)$$

自变量变化一个单位引起的优势比为exp(β)

$$\frac{\Omega^*}{\Omega}=\frac{P(y=1|x_1=1,x_2)}{1-P(y=1|x_1=1,x_2)}\Big/\frac{P(y=1|x_1=0,x_2)}{1-P(y=1|x_1=0,x_2)}$$

$$=\frac{P(y=1|x_1=1,x_2)}{P(y=1|x_1=0,x_2)}\times\frac{1-P(y=1|x_1=0,x_2)}{1-P(y=1|x_1=1,x_2)}=e^{\beta_1}$$

接近0时

$$\approx\frac{P(y=1|x_1=1,x_2)}{P(y=1|x_1=0,x_2)}$$

自变量变化一个单位引起响应概率比(相对风险)近似为exp(β)

- Logit模型示例：分析受教育程度对职业妇女外出工作的影响

```
> mroz_logit<-glm(INLF~NWIFEINC+EDUC+EXPER+EXPERSQ+AGE+KIDSLT6+KIDSGE6,data=Data,
+     family=binomial(link="logit"))
> summary(mroz_logit)

Call:
glm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE +
    KIDSLT6 + KIDSGE6, family = binomial(link = "logit"), data = Data)

Deviance Residuals:
    Min        1Q    Median        3Q       Max
-2.1770   -0.9063    0.4473    0.8561    2.4032

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.425452   0.860365   0.495  0.62095
NWIFEINC    -0.021345   0.008421  -2.535  0.01126 *
EDUC         0.221170   0.043439   5.091 3.55e-07 ***
EXPER        0.205870   0.032057   6.422 1.34e-10 ***
EXPERSQ     -0.003154   0.001016  -3.104  0.00191 **
AGE         -0.088024   0.014573  -6.040 1.54e-09 ***
KIDSLT6     -1.443354   0.203583  -7.090 1.34e-12 ***
KIDSGE6      0.060112   0.074789   0.804  0.42154
---
```

```
. logit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC, or nolog

Logistic regression                             Number of obs   =        753
                                                LR chi2(7)      =     226.22
                                                Prob > chi2     =     0.0000
Log likelihood = -401.76515                     Pseudo R2       =     0.2197
```

INLF	Odds Ratio	Std. Err.	z	P>|z|	[95% Conf.	Interval]
AGE	.9157386	.0133451	-6.04	0.000	.8899527	.9422715
EDUC	1.247536	.0541925	5.09	0.000	1.145717	1.358404
EXPER	1.228593	.0393849	6.42	0.000	1.153775	1.308263
EXPERSQ	.9968509	.0010129	-3.10	0.002	.9948676	.9988381
KIDSGE6	1.061956	.0794234	0.80	0.422	.9171603	1.22961
KIDSLT6	.2361344	.0480734	-7.09	0.000	.158441	.3519257
NWIFEINC	.978881	.0082436	-2.53	0.011	.9628565	.9951723
_cons	1.530283	1.316609	0.49	0.621	.2834155	8.262655

Note: _cons estimates baseline odds.

```
> exp(coef(mroz_logit))
(Intercept)    NWIFEINC        EDUC       EXPER     EXPERSQ         AGE     KIDSLT6
  1.5302825   0.9788810   1.2475360   1.2285929   0.9968509   0.9157386   0.2361344
    KIDSGE6
  1.0619557
```

妇女受教育年份增加一年，平均外出工作的概率是原来的exp(0.22)=1.25倍(倍数是常数)。可避开计算比例因子

二值响应模型的评价

- 评价指标：基于混淆矩阵计算：各类别错判率、总错判率

二分类预测的混淆矩阵

实际类别	预测类别	
	1	0
1	TP	FN
0	FP	TN

例：分析影响妇女外出工作的因素

$$P(y_i=1|\mathbf{x})>0.5, \hat{y}_i=1$$
$$P(y_i=1|\mathbf{x})\le 0.5, \hat{y}_i=0$$

```
> Porb<-predict(mroz_logit,type="response")  #Probility of y=1
> YesNot<-ifelse(Porb>0.5,1,0)
> (ConfuseMatrix<-table(Data$INLF,YesNot))
   YesNot
      0   1
  0 207 118
  1  81 347
> prop.table(ConfuseMatrix,1)*100
   YesNot
            0        1
  0 63.69231 36.30769
  1 18.92523 81.07477
> (err<-(sum(ConfuseMatrix)-sum(diag(ConfuseMatrix)))/sum(ConfuseMatrix))
[1] 0.2642762
```

- 总错判率的不足：受样本分布影响，尤其在非平衡样本中不适用

预测值

实际值		0	1	合计	正确率
	0	140	20	160	87.50%
	1	40	0	40	0%
	合计	180	20	200	70%

- 派生了其他评价指标，如敏感性(TPR)、特异性(TNR)、F1 score等

二值响应模型的评价

- 评价指标：麦克法登(McFadden)伪R^2(pseudo R-squared)

$$1-\frac{LL}{LL_0}$$

$$LL(y\,|\,\mathbf{x};\boldsymbol{\beta})=\sum_{i=1}^{n}[y_i\ln G(\mathbf{x_i}\boldsymbol{\beta})+(1-y_i)\ln(1-G(\mathbf{x_i}\boldsymbol{\beta}))]$$
$$=\sum_{i=1}^{n}[y_i\ln\frac{G(\mathbf{x_i}\boldsymbol{\beta})}{1-G(\mathbf{x_i}\boldsymbol{\beta})}+\ln(1-G(\mathbf{x_i}\boldsymbol{\beta}))]$$

$$\mathrm{P}(y_i=1\,|\,\mathbf{x}_i;\boldsymbol{\beta})=G(\mathbf{x}_i\boldsymbol{\beta})$$

- 分子为当前模型的对数似然函数值，分母是常数(零)模型的对数似然函数值
  - 对数似然函数值越大，表明所在模型参数下得到特定样本的可能性越大，模型的拟和优度高；反之；
- 若对数似然比与1无显著差异，说明解释变量全体对有效提高模型对样本的拟合无显著贡献；若对数似然比接近0，伪R^2接近1，说明解释变量全体有显著贡献

二值响应模型的评价

- 评价指标：麦克法登(McFadden)伪R^2(pseudo R-squared)

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$$LL(y\mid\mathbf{x};\boldsymbol{\beta})=\sum_{i=1}^{n}[y_i\ln G(\mathbf{x}_i\boldsymbol{\beta})+(1-y_i)\ln(1-G(\mathbf{x}_i\boldsymbol{\beta}))]$$
$$=\sum_{i=1}^{n}[y_i\ln\frac{G(\mathbf{x}_i\boldsymbol{\beta})}{1-G(\mathbf{x}_i\boldsymbol{\beta})}+\ln(1-G(\mathbf{x}_i\boldsymbol{\beta}))]$$

```
. logit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC, or nolog

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_cons	1.530283	1.316609	0.49	0.621	.2834155	8.262655

Note: _cons estimates baseline odds.

- McFadden伪R^2示例：

```
> mroz_logit0<-update(mroz_logit,formula=.~1)   #常数模型的对数似然
> (pseudoR2<-1-as.vector(logLik(mroz_logit)/logLik(mroz_logit0)))
[1] 0.2196814
```

二值响应模型的评价

- 模型的检验
  - 依据对数似然比，检验解释变量及解释变量全体是否显著提高了模型对样本的拟合程度
- 对数似然比：$\frac{LL}{LL_x}$
  - LL是解释变量x未进入模型时的对数似然值
  - LL_x是解释变量x进入模型后的对数似然值
  - 若：$\frac{LL}{LL_x}$与1无显著差异，表明解释变量x进入未显著提高对数似然值
  - 若：$\frac{LL}{LL_x} \gg 1$，表明解释变量x进入显著提高了对数似然值

模型的检验

- 似然比卡方统计量：$-\log(\frac{L}{L_x})^2 = -2\log(\frac{L}{L_x}) = -2\log(L) - (-2\log(L_x)) = -2LL - (-2LL_x)$
  - 服从自由度为1的卡方分布

- L和L_x是x引入回归方程前后的似然值
- 似然比卡方也称为模型偏差(deviance)
- $-2LL$、$-2LL_x$称为剩余偏差(residual deviance)

模型引入有效解释变量时将导致一个大的模型偏差(大的似然比卡方)，且当前的剩余偏差应较低

```
> anova(mroz_logit,test="Chisq")
Analysis of Deviance Table

Model: binomial, link: logit

Response: INLF

Terms added sequentially (first to last)


         Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                       752    1029.75
NWIFEINC  1   10.438       751    1019.31 0.0012347 **
EDUC      1   41.657       750     977.65 1.088e-10 ***
EXPER     1   80.692       749     896.96 < 2.2e-16 ***
EXPERSQ   1   13.716       748     883.24 0.0002126 ***
AGE       1   17.690       747     865.55 2.600e-05 ***
KIDSLT6   1   61.374       746     804.18 4.719e-15 ***
KIDSGE6   1    0.648       745     803.53 0.4208184
```

解释变量进入模型的顺序依数据框中变量的排列顺序

```
> 1-pchisq(0.648,1)
[1] 0.4208286
```

```
> 1-anova(mroz_logit)[8,4]/anova(mroz_logit)[1,4]
[1] 0.2196814
```

伪R方：$1-\frac{LL}{LL_0} = 1-\frac{-2LL}{-2LL_0}$

其他问题：完全分割或准完全分割

- 例：为研究美国各州执行死刑受哪些因素的影响，收集了美国不同地区(southern，是否南部地区yes/no)1946-1950的死刑率(excutions)，截至1951年在押刑事犯的服刑月数(time)，1949年的平均收入(income)，1950年劳动力参与率(lfp)，非白种人人口比例(noncauc)。建立logit模型

```
> murder_logit<-glm(I(executions>0)~time+income+noncauc+lfp+southern,data=MurderRates,
+   family=binomial(link="logit"))
Warning message:
glm.fit: fitted probabilities numerically 0 or 1 occurred
> table(I(MurderRates$executions>0),MurderRates$southern)

        no yes
  FALSE  9   0
  TRUE  20  15
```

```
Coefficients:
             Estimate  Std. Error z value Pr(>|z|)
(Intercept) 10.99326   20.77336   0.529   0.5967
time         0.01943    0.01040   1.868   0.0617 .
income      10.61013    5.65409   1.877   0.0606 .
noncauc     70.98785   36.41181   1.950   0.0512 .
lfp         -0.66763    0.47668  -1.401   0.1613
southernyes 17.33126 2872.17069   0.006   0.9952
```

- 15个南部地区全部执行死刑；29个非南部地区有部分执行
- 可以部分地依据地区来判断二值响应值

其他问题：完全分割或准完全分割

- 完全分割(Complete separation)和准完全分割(Quasi-complete separation)
  - 完全分割:依据解释变量x的取值，可确定全部观测的二值响应值取1还是0
  - 准完全分割:依据解释变量x的取值，可确定某类观测的二值响应值取1(或0)(如前例中的地区)
- 完全分割(或准完全分割)下，潜变量y^*的回归线的斜率为正无穷(或近似正无穷)
- 应首先做完全分割判断，注意模型提示信息，剔除相关自变量后建模

其他问题：Probit中的自变量遗漏问题

- 自变量遗漏：遗漏的变量c与$\mathbf{x}$不相关（可视c为不可观测的异质性）
  - 潜变量模型： $y^* = \mathbf{x}\boldsymbol{\beta} + \gamma c + e$ $y = 1[y^* > 0]$ and $e \mid \mathbf{x}, c \sim \text{Normal}(0, 1)$
  - 假设：$c \sim \text{Normal}(0, \tau^2)$

Probit模型

$\gamma c + e$, is independent of $\mathbf{x}$ 服从 $\text{Normal}(0, \gamma^2\tau^2 + 1)$ 非标准正态分布

$$\text{P}(y = 1 \mid \mathbf{x}) = \text{P}(\gamma c + e > -\mathbf{x}\boldsymbol{\beta} \mid \mathbf{x}) = \Phi(\mathbf{x}\boldsymbol{\beta}/\sigma) \quad \sigma^2 \equiv \gamma^2\tau^2 + 1$$

转成标准正态分布

  - 极大似然估计给出的一致性估计为：$\boldsymbol{\beta}/\sigma$

$$L(y \mid \mathbf{x}; \boldsymbol{\beta}) = \prod_{i=1}^{n} [G(\mathbf{x}_i\boldsymbol{\beta})]^{y_i} [1 - G(\mathbf{x}_i\boldsymbol{\beta})]^{1-y_i}$$

x_j的系数估计为：

$$\text{plim } \hat{\beta}_j = \beta_j/\sigma.$$

  - 因为：

$$\sigma = (\gamma^2\tau^2 + 1)^{1/2} > 1 \quad |\beta_j/\sigma| < |\beta_j|.$$

    - 不是β_j的一致性估计，存在向0的偏误

- 自变量遗漏：遗漏的变量c与$\mathbf{x}$不相关

$$\mathrm{P}(y=1\,|\,\mathbf{x})=\mathrm{P}(\gamma c+e>-\mathbf{x}\boldsymbol{\beta}\,|\,\mathbf{x})=\Phi(\mathbf{x}\boldsymbol{\beta}/\sigma) \quad \sigma^2\equiv\gamma^2\tau^2+1$$

  - 存在自变量遗漏下，x_j的偏效应估计值为：

$$(\beta_j/\sigma)\phi(\mathbf{x}\boldsymbol{\beta}/\sigma)$$

  - 不存在自变量遗漏下， x_j的偏效应估计值为：

$$\partial\mathrm{P}(y=1\,|\,\mathbf{x},c)/\partial x_j=\beta_j\phi(\mathbf{x}\boldsymbol{\beta}+\gamma c)$$

  - 所得不是效应的一致性估计，但：
    - 平均偏效应估计是可以接受的：虽第一部分小但第二部分可能大
    - 偏效应符号没有变化
    - 不影响两变量偏效应的对比(比值)：

$$(\beta_j/\sigma)/(\beta_h/\sigma)=\beta_j/\beta_h$$

其他问题：Probit中的连续内生变量问题

- Probit模型中连续内生解释变量(称存在不可观测的异质性)的偏效应估计

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + u_1$$

$$y_2 = \mathbf{z}_1\boldsymbol{\delta}_{21} + \mathbf{z}_2\boldsymbol{\delta}_{22} + v_2 = \mathbf{z}\boldsymbol{\delta}_2 + v_2$$

$$y_1 = 1[y_1^* > 0]$$

- 假定u_1服从标准正态分布
- y_2是内生的：u_1和v_2相关
  - 方程2为y_2的约简型
  - 假定v_2服从正态分布，给定$\mathbf{z}$时y_2服从正态分布(y_2连续)
  - u_1、v_2联合正态
- 自变量遗漏，可拓展到联立方程模型

- 利用关于潜变量的豪斯曼检验方程检验y_2的内生性并得到控制可能的内生性下的估计：

$$u_1 = \theta_1 v_2 + e_1$$ e_1和v_2不相关

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1$$ Probit估计需要关注e_1的分布

$$\theta_1 = \eta_1/\tau_2^2,\ \eta_1 = \mathrm{Cov}(v_2, u_1),\ \tau_2^2 = \mathrm{Var}(v_2),\quad \boldsymbol{Var(u_1) = \theta_1^2 Var(v_2) + Var(e_1) = \frac{\eta_1^2}{\tau_2^2} + Var(e_1)}$$

$$\mathrm{E}(e_1) = 0 \ \text{ and } \ \mathrm{Var}(e_1) = \mathrm{Var}(u_1) - \eta_1^2/\tau_2^2 = 1 - \rho_1^2$$

$$\rho_1 = \mathrm{Corr}(v_2, u_1)$$

- 所以有：

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1$$

$$e_1 \mid \mathbf{z}, y_2, v_2 \sim \mathrm{Normal}(0, 1 - \rho_1^2)$$

- Probit模型中的连续内生解释变量的偏效应估计

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + u_1$$
$$y_2 = \mathbf{z}_1\boldsymbol{\delta}_{21} + \mathbf{z}_2\boldsymbol{\delta}_{22} + v_2 = \mathbf{z}\boldsymbol{\delta}_2 + v_2$$
$$y_1 = 1[y_1^* > 0]$$

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1$$
$$e_1 \mid \mathbf{z}, y_2, v_2 \sim \text{Normal}(0, 1-\rho_1^2)$$

- 两步法估计
  - 第1:估计y_2的约简方程:y_2对z做OLS估计,得到残差$\hat{v}_2$
    - 以及: $\tau_2^2 = \text{Var}(v_2)$, τ_2^2的估计值$\widehat{\tau_2^2}$
  - 第2:将第1步得到的结果代入豪斯曼检验方程:
    - 建立Probit模型 $y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1$

$$\text{P}(y_1 = 1 \mid \mathbf{z}, y_2, v_2) = \Phi[(\mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2)/(1-\rho_1^2)^{1/2}]$$

e_1的标准差

    - 得到的估计值，记为:$\widehat{\boldsymbol{\delta}_{\rho1}}$, $\widehat{\alpha_{\rho1}}$, $\widehat{\theta_{\rho1}}$ 经过缩放的系数，大于未经缩放的系数(除非$\rho_1 = 0$)

$$\boldsymbol{\delta}_{\rho1} \equiv \boldsymbol{\delta}_1/(1-\rho_1^2)^{1/2} \quad \alpha_{\rho1} \equiv \alpha_1/(1-\rho_1^2)^{1/2} \quad \theta_{\rho1} \equiv \theta_1/(1-\rho_1^2)^{1/2}$$

其他问题：Probit中的连续内生变量问题

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + u_1$$

$$y_2 = \mathbf{z}_1\boldsymbol{\delta}_{21} + \mathbf{z}_2\boldsymbol{\delta}_{22} + v_2 = \mathbf{z}\boldsymbol{\delta}_2 + v_2$$

$$y_1 = 1[y_1^* > 0]$$

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1$$

$$e_1 \mid \mathbf{z}, y_2, v_2 \sim \text{Normal}(0, 1 - \rho_1^2)$$

- 又因为：

$$\frac{1}{1-\rho_1^2} = 1 + \theta_{\rho 1}^2 \tau_2^2$$

右边参数均已估计出来

- 可依据下式计算出参数δ_1, α_1的估计值(也可估计出θ_1)

$$\boldsymbol{\delta}_{\rho 1} \equiv \boldsymbol{\delta}_1/(1-\rho_1^2)^{1/2} \qquad \alpha_{\rho 1} \equiv \alpha_1/(1-\rho_1^2)^{1/2} \qquad \theta_{\rho 1} \equiv \theta_1/(1-\rho_1^2)^{1/2}$$

$$\hat{\boldsymbol{\delta}}_{\theta 1} \equiv \hat{\boldsymbol{\delta}}_{\rho 1}/(\hat{\theta}_{\rho 1}^2 \hat{\tau}_2^2 + 1)^{1/2} \qquad \hat{\alpha}_{\theta 1} \equiv \hat{\alpha}_{\rho 1}/(\hat{\theta}_{\rho 1}^2 \hat{\tau}_2^2 + 1)^{1/2}$$

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + u_1$$
$$y_2 = \mathbf{z}_1\boldsymbol{\delta}_{21} + \mathbf{z}_2\boldsymbol{\delta}_{22} + v_2 = \mathbf{z}\boldsymbol{\delta}_2 + v_2$$
$$y_1 = 1[y_1^* > 0]$$

$$y_1^* = \mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1$$
$$e_1 \mid \mathbf{z}, y_2, v_2 \sim \text{Normal}(0, 1-\rho_1^2)$$

- 进一步，考察y_2不同取值对y_1预测结果的影响：
  - 平均意义下：做差结果为(-1,0,1)

$$1[\bar{\mathbf{z}}_1\boldsymbol{\delta}_1 + \alpha_1(\bar{y}_2 + 1) + u_1 \geq 0] - 1[\bar{\mathbf{z}}_1\boldsymbol{\delta}_1 + \alpha_1\bar{y}_2 + u_1 \geq 0]$$

  - 方法1：基于原结构方程
    - 概率之差为： $\Phi[\bar{\mathbf{z}}_1\boldsymbol{\delta}_1 + \alpha_1(\bar{y}_2 + 1)] - \Phi(\bar{\mathbf{z}}_1\boldsymbol{\delta}_1 + \alpha_1\bar{y}_2)$
    - 按之前讨论的： $\Delta\hat{\mathrm{P}}(y = 1|\boldsymbol{x}) \approx [g(\hat{\beta}_0 + \boldsymbol{x}\hat{\boldsymbol{\beta}})\hat{\beta}_j]\Delta x_j,$
      - APE中的比例因子： $\frac{1}{n}\sum_{i=1}^{n}\phi(\mathbf{z}_{i1}\widehat{\boldsymbol{\delta}_{\theta 1}} + \widehat{\alpha_{\theta 1}}\, y_{2i})$
  - 方法2：基于第2步方程，计算APE中的比例因子(基于缩放的系数计算)

$$\frac{1}{n}\sum_{i=1}^{n}\phi(\mathbf{z}_{i1}\widehat{\boldsymbol{\delta}_{\rho 1}} + \widehat{\alpha_{\rho 1}}\, y_{2i} + \widehat{\theta_{\rho 1}}\widehat{v_{i2}})$$

- 例：研究丈夫收入对已婚妇女外出工作的影响

LPM, Logit, and Probit Estimates of Labor Force Participation

Dependent Variable: *inlf*

Independent Variable	LPM (OLS)	Logit (MLE)	Probit (MLE)
nwifeinc	−.0034 (.0015)	−.021 (.008)	−.012 (.005)
educ	.038 (.007)	.221 (.043)	.131 (.025)
exper	.039 (.006)	.206 (.032)	.123 (.019)
exper^2	−.00060 (.00019)	−.0032 (.0010)	−.0019 (.0006)
age	−.016 (.002)	−.088 (.015)	−.053 (.008)
kidslt6	−.262 (.032)	−1.443 (0.204)	−.868 (.119)
kidsge6	.013 (.013)	.060 (.075)	.036 (.043)
constant	.586 (.151)	.425 (.860)	.270 (.509)
Number of observations	753	753	753
Percent correctly predicted	73.4	73.6	73.4
Log-likelihood value	—	−401.77	−401.30
Pseudo *R*-squared	.264	.220	.221

```
. probit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 NWIFEINC,nolog

Probit regression                               Number of obs   =        753
                                                LR chi2(7)      =     227.14
                                                Prob > chi2     =     0.0000
Log likelihood = -401.30219                     Pseudo R2       =     0.2206

------------------------------------------------------------------------------
        INLF |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         AGE |  -.0528527   .0084772    -6.23   0.000    -.0694678   -.0362376
        EDUC |   .1309047   .0252542     5.18   0.000     .0814074     .180402
       EXPER |   .1233476   .0187164     6.59   0.000     .0866641    .1600311
     EXPERSQ |  -.0018871      .0006    -3.15   0.002     -.003063   -.0007111
     KIDSGE6 |    .036005   .0434768     0.83   0.408     -.049208    .1212179
     KIDSLT6 |  -.8683285   .1185223    -7.33   0.000    -1.100628    -.636029
    NWIFEINC |  -.0120237   .0048398    -2.48   0.013    -.0215096   -.0025378
       _cons |   .2700768    .508593     0.53   0.595    -.7267473    1.266901
------------------------------------------------------------------------------
```

- 怀疑nwifeinc是内生的

- 例：研究丈夫收入对已婚妇女外出工作的影响
  - NWIFEINC的内生性检验，并利用豪斯曼方程实现控制内生下的参数估计
    - nwifeinc的约简型方程，HUSEDUC为被排除的外生变量

```
> fit<-lm(NWIFEINC~HUSEDUC+EDUC+EXPER+EXPERSQ+AGE+KIDSLT6+KIDSGE6,data=Data)
> coeftest(fit,vcov=vcovHC(fit,type="HC0"))

t test of coefficients:

              Estimate  Std. Error t value   Pr(>|t|)
(Intercept) -1.4720e+01 3.5968e+00 -4.0926 4.731e-05 ***
HUSEDUC      1.1782e+00 1.6929e-01  6.9595 7.494e-12 ***
EDUC         6.7470e-01 2.5022e-01  2.6964  0.007167 **
EXPER       -3.1299e-01 1.3543e-01 -2.3110  0.021105 *
EXPERSQ     -4.7756e-04 3.9653e-03 -0.1204  0.904170
AGE          3.4015e-01 5.3798e-02  6.3227 4.426e-10 ***
KIDSLT6      8.2627e-01 1.0386e+00  0.7956  0.426527
KIDSGE6      4.3553e-01 2.8706e-01  1.5172  0.129635
---
```

$$P(y_1 = 1 \mid \mathbf{z}, y_2, v_2) = \Phi[(\mathbf{z}_1\boldsymbol{\delta}_1 + \alpha_1 y_2 + \theta_1 v_2)/(1-\rho_1^2)^{1/2}]$$

```
> mroz_probit<-glm(INLF~NWIFEINC+EDUC+EXPER+EXPERSQ+AGE+KIDSLT6+KIDSGE6+fit$residuals,data=Data,
+     family=binomial(link="probit"))
> summary(mroz_probit)

Call:
glm(formula = INLF ~ NWIFEINC + EDUC + EXPER + EXPERSQ + AGE + 
    KIDSLT6 + KIDSGE6 + fit$residuals, family = binomial(link = "probit"), 
    data = Data)

Deviance Residuals: 
    Min       1Q   Median       3Q      Max  
-2.2523  -0.9078   0.4204   0.8566   2.2803  

Coefficients:
               Estimate Std. Error z value Pr(>|z|)    
(Intercept)   0.0171183  0.5380339   0.032  0.97462    
NWIFEINC     -0.0368639  0.0183848  -2.005  0.04495 *  
EDUC          0.1702142  0.0377615   4.508 6.56e-06 ***
EXPER         0.1163118  0.0193869   6.000 1.98e-09 ***
EXPERSQ      -0.0019458  0.0005999  -3.244  0.00118 ** 
AGE          -0.0449529  0.0101351  -4.435 9.19e-06 ***
KIDSLT6      -0.8444319  0.1197268  -7.053 1.75e-12 ***
KIDSGE6       0.0477912  0.0449431   1.063  0.28761    
fit$residuals 0.0267092  0.0191539   1.394  0.16318    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

得到的是经过缩放的系数估计

$$\boldsymbol{\delta}_{\rho 1} \equiv \boldsymbol{\delta}_1/(1-\rho_1^2)^{1/2}$$

$$\rho_1 = \mathrm{Corr}(v_2, u_1)$$

无法拒绝外生假设(内生性不显著)：$\rho_1 = 0$

- 例：研究丈夫收入对已婚妇女外出工作的影响
  - NWIFEINC的内生性检验，并利用豪斯曼方程实现控制内生下的参数估计
    - nwifeinc的约简型方程，HUSEDUC为被排除的外生变量

$$\delta_{\rho 1} \equiv \delta_1/(1-\rho_1^2)^{1/2}$$

```
. ivprobit INLF AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 (NWIFEINC = HUSEDUC),first twostep
Checking reduced-form model...
first-stage regression
```

若略去twostep则采用IV Probit估计

Source	SS	df	MS
Model	20676.7705	7	2953.82435
Residual	81120.3451	745	108.886369
Total	101797.116	752	135.368505

Number of obs = 753
F(7, 745) = 27.13
Prob > F = 0.0000
R-squared = 0.2031
Adj R-squared = 0.1956
Root MSE = 10.435

NWIFEINC	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
HUSEDUC	1.178155	.1609449	7.32	0.000	.8621956	1.494115
AGE	.3401521	.0597084	5.70	0.000	.2229354	.4573687
EDUC	.6746951	.2136829	3.16	0.002	.2552029	1.094187
EXPER	-.3129877	.1382549	-2.26	0.024	-.5844034	-.0415721
EXPERSQ	-.0004776	.0045196	-0.11	0.916	-.0093501	.008395
KIDSGE6	.4355289	.3219888	1.35	0.177	-.1965845	1.067642
KIDSLT6	.8262719	.8183785	1.01	0.313	-.7803305	2.432874
_cons	-14.72048	3.787326	-3.89	0.000	-22.15559	-7.285383

Two-step probit with endogenous regressors

Number of obs = 753
Wald chi2(7) = 173.79
Prob > chi2 = 0.0000

	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
NWIFEINC	-.0368641	.0186314	-1.98	0.048	-.0733809	-.0003472
AGE	-.044953	.010327	-4.35	0.000	-.0651936	-.0247125
EDUC	.1702153	.0384014	4.43	0.000	.0949499	.2454806
EXPER	.1163123	.0197084	5.90	0.000	.0776846	.15494
EXPERSQ	-.0019459	.0006129	-3.17	0.001	-.0031471	-.0007446
KIDSGE6	.0477905	.045177	1.06	0.290	-.0407549	.1363359
KIDSLT6	-.8444363	.1218529	-6.93	0.000	-1.083264	-.605609
_cons	.0171187	.5498911	0.03	0.975	-1.060648	1.094885

```
Instrumented:  NWIFEINC
Instruments:   AGE EDUC EXPER EXPERSQ KIDSGE6 KIDSLT6 HUSEDUC
```

无法拒绝外生假设(内生性不显著)

LM统计量

```
Wald test of exogeneity: chi2(1) = 1.99          Prob > chi2 = 0.1584
```